

Summary of my Shopping Cart System

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BSIT 1-2

During the development of this program, I made several improvements to enhance its structure, readability, and overall functionality. The initial version already supported basic product listing, cart handling, and checkout, but I later refined the system to make the flow more organized and easier to follow. One of the improvements was structuring the product display using formatted output so that data appears in a table-like format in the console, such as:

```
Console.WriteLine($"{id,-3} | {name,-25} | {category,-18} | ₱{price,8:F2} | {remainingStock,5}");
```

This helped improve clarity when viewing products and cart items. I also improved the modularity of the program by separating concerns into methods like `DisplayProduct()`, `DisplayCartItem()`, and `DisplayCartHeader()`, which made the code easier to manage and reuse across different parts of the system.

Another major improvement was in input validation and user flow control. I refined multiple loops to ensure that the system only accepts valid inputs and handles empty or incorrect entries more effectively. For example, I improved the validation logic for menu selection and product searching using conditions like:

```
if (!int.TryParse(Console.ReadLine(), out menu))  
{  
    Console.WriteLine("Invalid Input.");  
    continue;  
}
```

This helped prevent runtime errors and improved the stability of the program during user interaction.

Lastly, I enhanced the checkout and transaction handling system. Stock management was made more reliable by properly updating inventory using methods such as:

```
product.remainingStock += cart[updateIndex].quantity;  
product.DeductStock(newQuantity);
```

I also added clearer output for receipts, order history, and low stock alerts to improve usability. Overall, these refinements helped transform the program into a more structured and functional console-based shopping system with better user experience and maintainability.

Screenshots of Initial Output:

Main Menu:

ID	Name	Category	Price	Stock
101	Instant Noodles Pack	Food	₹ 120.00	20
102	Chocolate Bar	Food	₹ 50.00	25
103	Bottled Water 1L	Food	₹ 30.00	30
201	Plain T-Shirt	Clothing	₹ 300.00	15
202	Jeans Pants	Clothing	₹ 900.00	8
203	Hoodie Jacket	Clothing	₹ 1200.00	6
301	Notebook	School Supplies	₹ 60.00	18
302	Ballpen (Blue)	School Supplies	₹ 15.00	50
303	Backpack	School Supplies	₹ 850.00	9

=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6):

Add a Product:

ID	Name	Category	Price	Stock
101	Instant Noodles Pack	Food	₹ 120.00	20
102	Chocolate Bar	Food	₹ 50.00	25
103	Bottled Water 1L	Food	₹ 30.00	30
201	Plain T-Shirt	Clothing	₹ 300.00	15
202	Jeans Pants	Clothing	₹ 900.00	8
203	Hoodie Jacket	Clothing	₹ 1200.00	6
301	Notebook	School Supplies	₹ 60.00	18
302	Ballpen (Blue)	School Supplies	₹ 15.00	50
303	Backpack	School Supplies	₹ 850.00	9

=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6): 1

=== ADD A PRODUCT ===
Enter Product ID: 101

Enter Quantity: 2

Item Successfully Added To Cart.

Do You Want To Add More Items?(Y/N): _

Cart:

ID	Name	Category	Price	Stock
101	Instant Noodles Pack	Food	? 120.00	20
102	Chocolate Bar	Food	? 50.00	25
103	Bottled Water 1L	Food	? 30.00	30
201	Plain T-Shirt	Clothing	? 300.00	15
202	Jeans Pants	Clothing	? 900.00	8
203	Hoodie Jacket	Clothing	? 1200.00	6
301	Notebook	School Supplies	? 60.00	18
302	Ballpen (Blue)	School Supplies	? 15.00	50
303	Backpack	School Supplies	? 850.00	9

=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6): 2

=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5):

View Cart:

=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5): 1

=== VIEW CART ===

No	Name	Qty	Price	Subtotal
1	Instant Noodles Pack	2	? 120.00	? 240.00

=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5):

Remove Cart Item:

=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5): 2

=== REMOVE CART ITEM ===

No	Name	Qty	Price	Subtotal
1	Instant Noodles Pack	2	? 120.00	? 240.00

Enter Item Number to Remove: 1

Item is Removed Successfully.

Update Quantity:

```
=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5): 3

=== UPDATE QUANTITY ===

No | Name | Qty | Price | Subtotal
-----
1 | Instant Noodles Pack | 2 | ? 120.00 | ? 240.00

Enter Item Number To Update: 1
Enter New Quantity: 5
Quantity Updated.
```

Clear Cart:

```
=== CART MENU ===
1. View Cart
2. Remove Cart Item
3. Update Quantity
4. Clear Cart
5. Back
Enter Choice(1-5): 4

=== CLEAR CART ===
Are you sure you want to remove all items in your cart?(Y/N): y
Cart cleared successfully.
```

Search Product Name:

```
=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6): 3

=== SEARCH PRODUCT NAME ===
Enter Product Name to Search: jacket

ID | Name | Category | Price | Stock
-----
203 | Hoodie Jacket | Clothing | ? 1200.00 | 6
```

Search Product Category:

```
=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6): 4

=== SEARCH PRODUCT CATEGORY ===
1. Food
2. Clothing
3. School Supplies
4. Back
Enter Choice (1-4): 1

ID | Name | Category | Price | Stock
-----
101 | Instant Noodles Pack | Food | ? 120.00 | 20
102 | Chocolate Bar | Food | ? 50.00 | 25
103 | Bottled Water 1L | Food | ? 30.00 | 30
```

Checkout:

```
=== MAIN MENU ===
1. Add a Product
2. Cart
3. Search Product Name
4. Search Product Category
5. Checkout
6. Exit
Enter Choice (1-6): 5

=== CHECKOUT ===

No  | Name                | Qty | Price | Subtotal
-----
1   | Instant Noodles Pack | 2   | ? 120.00 | ? 240.00
2   | Hoodie Jacket        | 6   | ? 1200.00 | ? 7200.00

Grand Total: ?7440.00
Discount: ?744.00
Final Total: ?6696.00
Enter Payment: 7000

=== RECEIPT ===
Receipt No: 0001
Date: 01/05/2026 1:57:54 am
x2 Instant Noodles Pack = ?240.00
x6 Hoodie Jacket = ?7200.00
Grand Total: ?7440.00
Discount: ?744.00
Final Total: ?6696.00
Payment: ?7000.00
Change: ?304.00

=== LOW STOCK ALERT ===
Hoodie Jacket has no stock left.

=== ORDER HISTORY ===
Receipt #0001 - Final Total: ?6696.00 - Date: 01/05/2026 1:57:54 am

Checkout complete!

Do You Want To Shop Again?(Y/N):
```